

1. TITLE OF THE EXPERIMENT

Design and Implementation of a Styled Web Page Using CSS3 and an Interactive HTML5 Canvas Graphic with JavaScript – “My Styled Webpage”

2. AIM

To design and implement a web page that applies external CSS3 styling to HTML elements such as headers, navigation bars, cards and buttons, and to draw and render shapes and text on an HTML5 <canvas> element using JavaScript.

3. ALGORITHM / DESIGN

- Step 1:** Create the basic HTML document skeleton using <html>, <head>, <title> and <body> tags, and link an external stylesheet using <link rel="stylesheet" href="style.css">.
- Step 2:** Design the page header using <header>, <h1> and a navigation bar using <nav> with <a> hyperlinks.
- Step 3:** Add an HTML5 <canvas> element with a defined width and height to serve as a drawing surface.
- Step 4:** Create a <section> with class "container" holding multiple <div class="card"> elements, each containing a heading, paragraph and a <button>.
- Step 5:** Add a <footer> with copyright information.
- Step 6:** Write a <script> block that accesses the canvas using getElementById() and obtains its 2D drawing context using getContext("2d").
- Step 7:** Use the context methods fillRect(), beginPath()/arc()/stroke() and moveTo()/lineTo()/stroke() to draw a rectangle, a circle and a line on the canvas.
- Step 8:** Set the font using ctx.font and render text on the canvas using ctx.fillText().
- Step 9:** Style the header, nav, cards and buttons in an external style.css file (layout, colors, spacing, hover effects).
- Step 10:** Save the HTML and CSS files and execute the page by opening it in a web browser.
- Step 11:** Observe and verify that the styling is applied correctly and the canvas graphic renders as expected.

4. TEST CASES & EXECUTION DATA

TC No.	Feature / Tag Tested	Expected Result	Status
TC-01	External CSS linking (<link>)	style.css rules apply to header, nav, cards and buttons on page load	Passed
TC-02	Navigation bar styling	Nav links display in a styled horizontal bar with defined spacing/colors	Passed
TC-03	Card layout (<div class="card">)	Three cards (HTML5, CSS3, JavaScript) render side by side with heading, text and button	Passed
TC-04	Canvas rectangle – ctx.fillRect()	A filled rectangle is drawn at (20,20) with width 120 and height 70	Passed
TC-05	Canvas circle – ctx.arc()	A stroked circle of radius 35 is drawn centered at (220,55)	Passed
TC-06	Canvas line – moveTo()/lineTo()	A straight line is drawn from (20,130) to (300,130)	Passed
TC-07	Canvas text – ctx.fillText()	The text “sai” renders in 28px Arial font at position (170,190)	Passed
TC-08	Read More buttons	Each card button renders and responds to click/hover as styled in CSS	Passed

5. LEARNING OUTCOME / RESULT

Successfully designed and implemented a styled web page that links an external CSS3 stylesheet for its header, navigation bar and card layout, and used JavaScript with the HTML5 Canvas API to draw a rectangle, a circle, a line and text at runtime. The output was verified in a web browser and matched the expected rendering for all test cases. This experiment strengthened the understanding of separating structure (HTML) from presentation (CSS) and using JavaScript to control the Canvas 2D rendering context for dynamic graphics.

6. PROGRAM / SOURCE CODE

File name: index.html

```
<!DOCTYPE html>
<html>
<head>
<title>My Styled Webpage</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h1>My Styled Webpage</h1>
</header>
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Contact</a>
</nav>
<canvas id="myCanvas" width="500" height="220"></canvas>
<section class="container">
<div class="card">
<h2>HTML5</h2>
<p>HTML is used to create web pages.</p>
<button>Read More</button>
</div>
<div class="card">
<h2>CSS3</h2>
<p>CSS is used to style web pages.</p>
<button>Read More</button>
</div>
<div class="card">
<h2>JavaScript</h2>
<p>JavaScript adds interactivity.</p>
<button>Read More</button>
</div>
</section>
<footer>
<p>&copy; 2026 ANITS</p>
</footer>
<script>
let canvas=document.getElementById("myCanvas");
let ctx=canvas.getContext("2d");
ctx.fillRect(20,20,120,70);
ctx.beginPath();
ctx.arc(220,55,35,0,2*Math.PI);
ctx.stroke();
ctx.beginPath();
ctx.moveTo(20,130);
ctx.lineTo(300,130);
ctx.stroke();
ctx.font="28px Arial";
ctx.fillText("sai",170,190);
</script>
</body>
</html>
```